

ANSWERS

EXERCISE 1.1

1. (i), (ii) and (vi) are sets 2. (i) $\in$ (ii) $\notin$ (iii) $\notin$ (iv) $\in$ (v) $\in$ (vi) $\notin$
3. (i) {Karachi, Lahore, Quetta, Peshawar, Gilgit}
(ii) {50, 51, 52, ..., 69, 70} (iii) {2, 4, 6, 8, 10, 12, 14, 16, 18, 20}
(iv) {p, a, k, i, s, t, n} (vi) {White, Green}
4. (i) True (ii) True (iii) False (iv) False (v) False
(vi) True (vii) False (viii) True

EXERCISE 1.2

1. (i) "0" is repeating (ii) "★" is repeating (iii) "a" is repeating
(iv) "2" and "3" and "4" are repeating
2. (i) Finite (ii) Infinite (iii) Infinite (iv) Finite (v) Finite
(vi) Infinite (vii) Infinite (viii) Finite (ix) Finite (x) Infinite
3. (i), (ii), (iv) and (v) are empty sets 4. (i) and (ii) are equal sets
5. (i), (ii) and (iii) are equivalent sets
6. (i) True (ii) False (iii) True (iv) False (v) True
7. (i) $\subseteq$ (ii) $=$ (iii) $\supseteq$ (iv) $\supseteq$ (v) $\sim$
(vi) $\subseteq$ (vii) $\subset$ (viii) $\subseteq$

REVIEW EXERCISE 1

1. (i) elements (ii) empty set (iii) all same
2. (i) d (ii) d (iii) d (iv) c
4. (i) $A = \{4, 6, 8\}$ (ii) $B = \{1, 3, 5, \dots, 15, 17\}$
(iii) $C = \{2, 4, 6, \dots, 26, 28\}$ (vi) $D = \{\text{January, June, July}\}$
5. (i) A is a finite set (ii) B is an infinite set (iii) C is a finite set
(iv) D is an infinite set (v) E is a finite set

6. F and M are empty sets

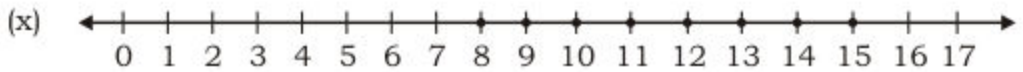
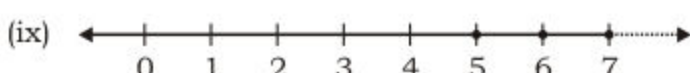
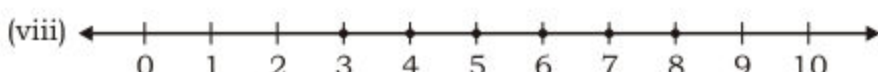
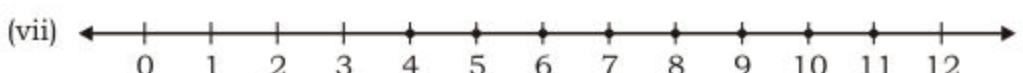
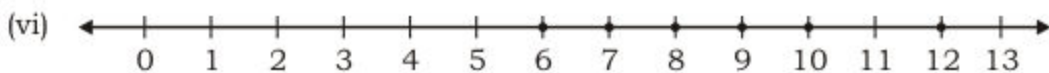
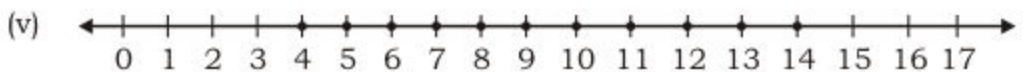
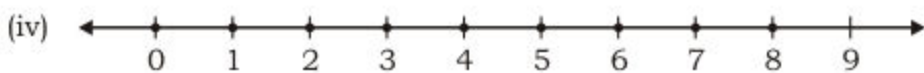
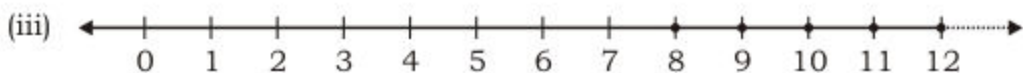
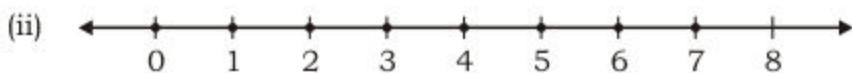
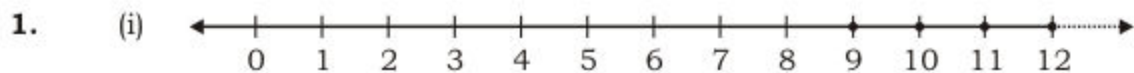
7. (i) True (ii) True (iii) False (iv) True

8. (i) True (ii) True (iii) False (iv) False (v) True
(vi) True (vii) False (viii) False (ix) False

EXERCISE 2.1

1. (i) 1 (ii) 0 (iii) Not possible (iv) Not possible

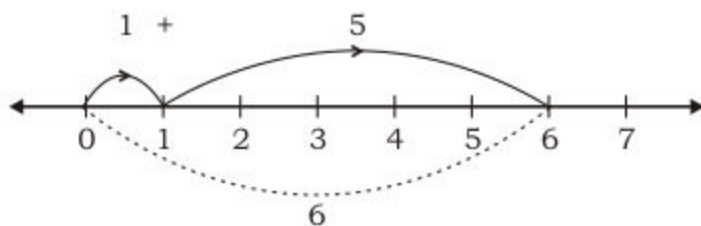
2. 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 3. 0, 1, 2, 3, 4, 5, 6, 7, 8, 9



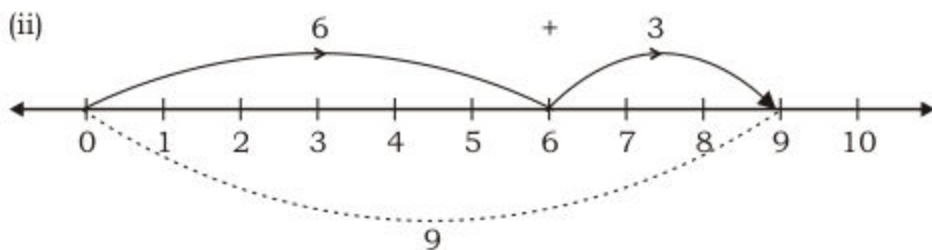
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5.

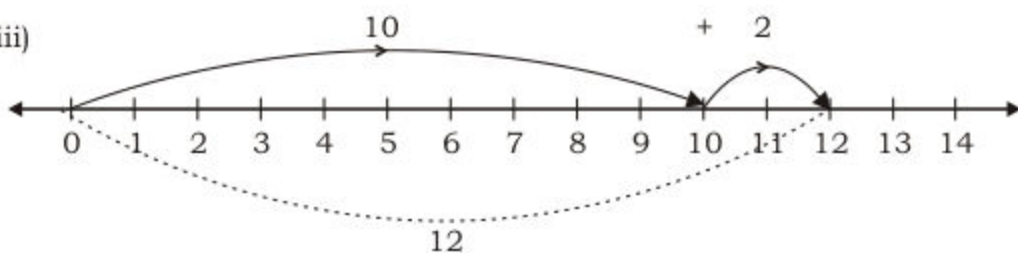
(i)



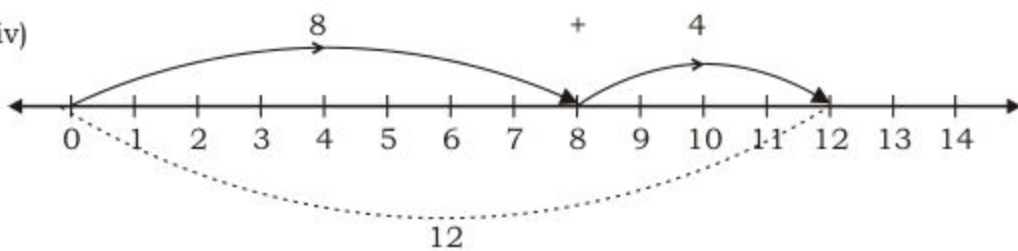
(ii)



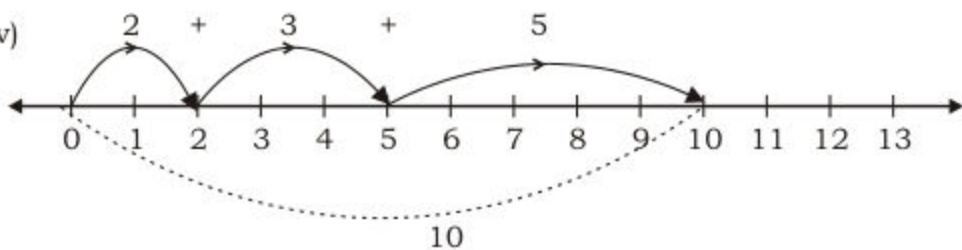
(iii)



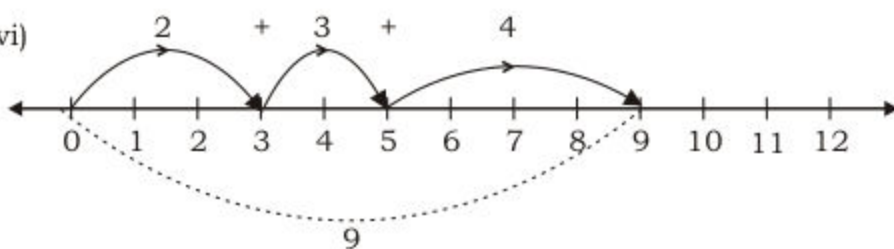
(iv)



(v)



(vi)



EXERCISE 2.2

1. Required sum is 1,999,999
2. Number of children is 715
3. Rs 42300 left in account
4. Rs 49615 left with him
7. (i) 849 (ii) 97864 (iii) 749,749 (iv) 430 (v) 4946
8. The starting number

EXERCISE 2.3

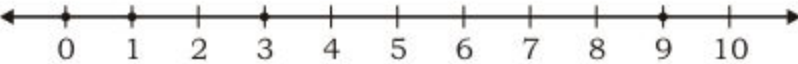
1. (i) 81984 (ii) 75808 (iii) 258048 (iv) 157210
2. (i) Quotient = 134, Remainder = 0 (ii) Quotient = 393, Remainder = 39
(iii) Quotient = 16, Remainder = 25 (iv) Quotient = 309, Remainder = 145
3. 30 trees in each row
4. Rs 1,227,500
5. Required number = 99975
6. 999900

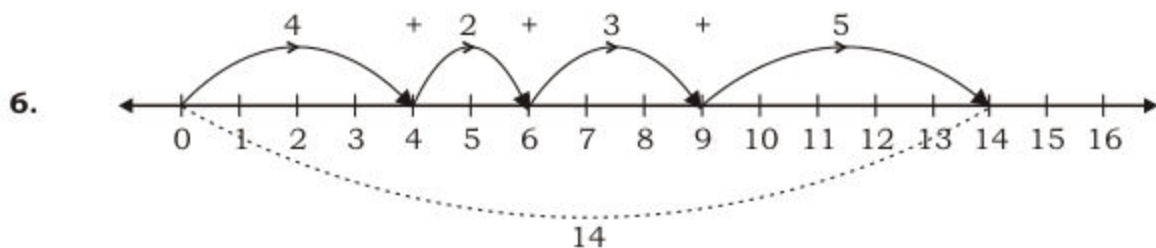
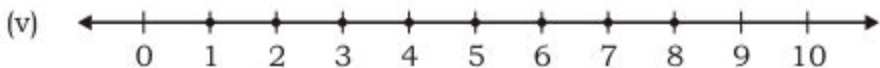
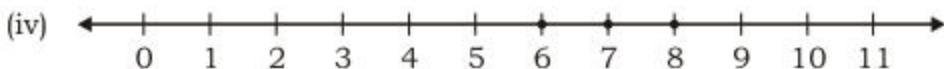
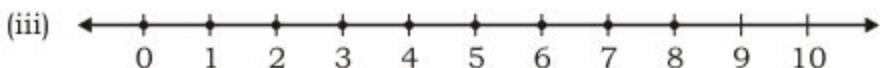
EXERCISE 2.4

1. (i) 0 (ii) 6 (iii) 2 (iv) 4 (v) 7, 5 (vi) 964 (vii) -, 125
2. (i) False (ii) False (iii) True (iv) True

REVIEW EXERCISE 2

1. (i) 3, 2 (ii) 528, 112 (iii) 8 (iv) 1, 7 (v) 1, 1
(vi) 0, 0 (vii) 9, 9 (viii) 20, 1 (ix) 8, 17 (x) 32, 32
2. (i) False (ii) False (iii) False (iv) True (v) True
(vi) False (vii) False (viii) True (ix) False (x) True
3. (i) Predessor = 670, Successor = 672 (ii) Predessor = 244, Successor = 246
4. (i) b (ii) a (iii) d (iv) b (v) b

5. (i) 
- (ii) 



11. Rs 960

12. Required number = 9988

EXERCISE 3.1

1. a and c
2. (a) 1, 3, 5, 15 (b) 1, 2, 3, 5, 6, 10, 15, 30 (c) 1, 2, 5, 10, 25, 50
(d) 1, 5, 25, 125 (e) 1, 2, 3, 5, 6, 10, 15, 25, 30, 50, 75, 150
3. b and c
4. (a) 4, 8, 12, 16, 20, 24, 28 (b) 7, 14, 21, 28, 35, 42, 49
(c) 12, 24, 36, 48, 60, 72, 84 (d) 15, 30, 45, 60, 75, 90, 105
5. (a) 7, 11 and 13 (b) 23, 29, 31 and 37
(c) 61, 67, 71, 73, 79, 83 and 89
6. 2 is an even prime number
7. (a) even (b) odd (c) odd (d) odd (e) even (f) odd
8. 22, 24, 26, 28, 30, 32, 34, ..., 48, 50
9. 11, 13, 15, 17, 19, ..., 37, 39
10. 4, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20, 21, 22, 24, 25, 26, 27, 28, 30, 32, 33, 34, 35, 36, 38, 39.
11. 72, 74, 75, 76, 77, 78, 80, 81, 82, 84, 85, 86, 87, 88, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100.
12. 23, 29, 31, 37, 41, 43, 47, 51, 53, 59.
13. Because 1 has only one factor.

EXERCISE 3.2

1. (i) Yes (ii) Yes (iii) Yes (iv) No (v) No (vi) Yes
 (vii) Yes (viii) No (ix) Yes (x) Yes (xi) Yes (xii) No
2. (ii) and (iii)
3. (ii) and (iii)
4. (i) and (ii)
5. (ii) and (iii)
6. (ii)
7. (i) and (iii)
8. (i) and (iii)
9. (i) and (iii)
10. (i) and (iii)
11. (i) and (iii)

EXERCISE 3.3

1. (i) $2^3 \times 3^2 \times 5^1$ (ii) $3^1 \times 5^3 \times 7^2$ (iii) $2^1 \times 3^4 \times 7^2$
2. (i) $2^3 \times 3^1$ (ii) $2^4 \times 3^1$ (iii) $2^3 \times 3^3$ (iv) $2^1 \times 5^3$
 (v) $2^2 \times 3^2 \times 13^1$ (vi) $2^2 \times 3^3 \times 5^1$ (vii) 2^{10} (viii) $2^3 \times 5^4$
3. Yes

EXERCISE 3.4

1. (i) 25 (ii) 98 (iii) 18 (iv) 12 (v) 53 (vi) 12 (vii) 10
2. (i) 4 (ii) 9 (iii) 55 (iv) 36 (v) 747 (vi) 30

EXERCISE 3.5

1. (i) 600 (ii) 441 (iii) 5760 (iv) 1620 (v) 6825 (vi) 144
2. (i) 495 (ii) 210 (iii) 120 (iv) 1080
3. LCM = 6

EXERCISE 3.6

1. Greatest Length = 180 cm
2. Greatest Number = 75
3. Largest Number = 16
4. Maximum Capacity = 30 litres
5. Longest Tape = 2 m
6. Required Time = 8.30 a.m.
7. 189
8. Least Quantity = 48 litres
9. Required Time = 7 minutes 12 seconds

REVIEW EXERCISE 3

1. (i) 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60,
(ii) 1, 2, 5, 10, 25, 50, 125, 250
2. (i) 13, 26, 39, 52, 65 (ii) 20, 40, 60, 80, 100
3. 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47
4. 32, 33, 34, 35, 36, 38, 39, 40, 42, 44, 45, 46, 48, 49, 50, 52, 54, 55, 56, 57, 58.
5. (ii), (iii) and (iv) 6. (i), (iii) and (iv) 7. (ii) and (iv)
8. (i) and (ii) 9. (ii) and (iii)
10. (i) $2^1 \times 3^2 \times 5^2$ (ii) $2^4 \times 3^2 \times 5^1$
11. (i) HCF = 6, LCM = 72 (ii) HCF = 1, LCM = 120
12. (i) HCF = 5, LCM = 1040 (ii) HCF = 13, LCM = 1690
13. Greatest Number = 10 14. 186

EXERCISE 4.1

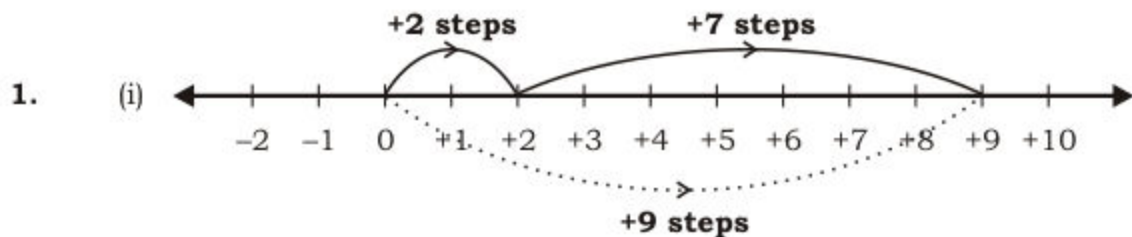
1. (i) 
- (ii) 
- (iii) 
- (iv) 
2. (i) $+15 > -6$ (ii) $-8 < 0$ (iii) $+16 > 0$ (iv) $-2 > -8$ (v) $+ > < +9$ (vi) $-4 < -1$
3. (i) Left (ii) Right (iii) Right (iv) Left
4. (i) $>$ (ii) $<$ (iii) $>$ (iv) $<$ (v) $<$ (vi) $>$
5. (i) Ascending order: -7, -3, -1, 0, +1, +5
Descending order: +5, +1, 0, -1, -3, -7
- (ii) Ascending order: -3, -1, 0, +2, +4, +5
Descending order: +5, +4, +2, 0, -1, -3

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- (iii) Ascending order: $-5, -4, 0, +4, +5$
Descending order: $+5, +4, 0, -4, -5$
- (iv) Ascending order: $-8, -7, -4, -2, -1$
Descending order: $-1, -2, -4, -7, -8$

6. (i) 5 (ii) 20 (iii) 0 (iv) 18 (v) 50
7. (i) Ascending order: $0, 1, 3, 4, 5, 6$
Descending order: $6, 5, 4, 3, 1, 0$
- (ii) Ascending order: $0, 10, 17, 25, 30, 60$
Descending order: $60, 30, 25, 17, 10, 0$
- (iii) Ascending order: $0, 4, 7, 10, 11, 20$
Descending order: $20, 11, 10, 7, 4, 0$
- (iv) Ascending order: $3, 5, 8, 9, 12, 13$
Descending order: $13, 12, 9, 8, 5, 3$
8. (i) True (ii) False (iii) True (iv) False
(v) False (vi) True (vii) True

EXERCISE 4.2



Students may draw number line in their own copies themselves

- (ii) -11 steps (iii) $+3$ steps (iv) -6 steps
(v) -3 steps (vi) 0 steps
2. (i) $+10$ steps (ii) -12 steps (iii) $+10$ steps (iv) -13 steps
3. (i) $+30$ (ii) -40 (iii) -27 (iv) -9
(v) -6 (vi) 0

EXERCISE 4.3

1. (i) $+2$ steps (ii) -5 steps (iii) -6 steps (iv) -14 steps
(v) $+11$ steps (vi) -11 steps

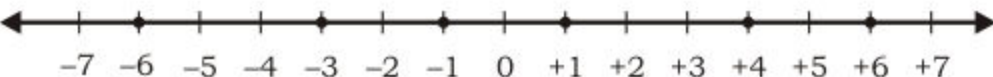
ANSWERS

2. (i) -16 (ii) -10 (iii) +1 (iv) +5 (v) +20 (vi) +3
3. (i) +40 (ii) -5 (iii) -16 (iv) +10 (v) +2 (vi) +35
(vii) -46 (viii) -50 (ix) -10

EXERCISE 4.4

1. (i) -60 (ii) +340 (iii) +400 (iv) -432 (v) -490 (vi) +688
2. (i) -5 (ii) +5 (iii) +25 (iv) -63 (v) -78 (vi) +21
3. (i) False (ii) True (iii) False (iv) True (v) False (vi) True
(vii) False (viii) True

REVIEW EXERCISE 4

1. -4, -3, -2, -1, 0, +1, +2, +3, +4
2. (i) 
(ii) 
3. (i) 3, 4, 4, 6, 0 (ii) 2, 7, 1, 0, 4, 2
4. (i) -3 steps (ii) -15 steps (iii) +11 steps (iv) 0 steps
5. (i) +2 steps (ii) -5 steps (iii) +7 steps
6. (i) -4 (ii) -14 (iii) -3 (iv) +30 (v) -5 (vi) +15
7. (i) -50 (ii) +54 (iii) +120 (iv) -45 (v) 0 (vi) +80
8. (i) -5 (ii) +5 (iii) -6 (iv) +4

EXERCISE 5.1

1. $\frac{7}{20}$ 2. $1\frac{43}{60}$ 3. $2\frac{33}{56}$ 4. $\frac{7}{8}$ 5. $\frac{1}{68}$
6. $9\frac{5}{6}$ 7. $-\frac{1}{2}$ 8. $3\frac{15}{22}$ 9. $12\frac{1}{4}$ 10. $14\frac{1}{6}$

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11. $10\frac{7}{12}$ 12. $1\frac{1}{30}$ 13. $1\frac{6}{49}$ 14. 28.944 15. 24.373
16. 53.01 17. 0.88 18. 2.408 19. 86.965 20. 8.207
21. 0.95 22. 4.04 23. 3.8

EXERCISE 5.2

1. 48 students 2. 70 oranges left 3. $\frac{19}{56}$ on foot
4. Rs 19500 left 5. $\frac{16}{17}$ required fraction 6. $\frac{16}{35}$ required fraction
7. $\frac{92}{265}$ required fraction 8. Rs 3023.50 total amount
9. $\frac{19}{100}$ required fraction 10. Rs 4926.55 required amount

REVIEW EXERCISE 5

1. (i) $\frac{21}{29}$ (ii) $\frac{9}{20}$ (iii) 23.951 (iv) 4.475
2. (i) $9\frac{1}{7}$ kg left (ii) $\frac{3}{5}$ solved and $\frac{2}{5}$ unsolved questions
3. $\frac{3^{\text{rd}}}{5}$ part on next day, 60 pages on 1st day, 90 pages on next day
4. Rs 135523.03 required amount 5. (i) True (ii) False (iii) True
6. (i) Multiplication (ii) Braces or curly brackets
(iii) Parenthesis, braces, square brackets

EXERCISE 6.1

1. (i) 2 : 9 (ii) 5 : 6 (iii) 1 : 75 (iv) 3 : 1 (v) $p : q$
2. (i) $\frac{1}{5}$ (ii) $\frac{2}{19}$ (iii) 8 (iv) $\frac{75}{76}$ (v) $\frac{x}{y}$
2. (i) 2 : 25 (ii) 1 : 90 (iii) 5 : 7 (iv) 3 : 28 (v) 25 : 9
(vi) 2 : 15 (vii) 5 : 8 (viii) 6 : 3 : 2 (ix) 15 : 50 : 58 (x) 34 : 6 : 35
(xi) 6 : 3 : 2 (xii) 6 : 3 : 21

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3. (i) 5 : 6 (ii) 5 : 2 (iii) 7 : 10 (iv) 6 : 7 (v) 20 : 1
(vi) 15 : 14 : 30
4. 5 : 9 5. 4 : 3 6. 2 : 5 7. 10 : 9 8. 1 : 2 : 3

EXERCISE 6.2

1. (i) means = 5 and 8, extremes = 2 and 20
(ii) means = 4 and 6, extremes = 3 and 8
(iii) means = b and c , extremes = a and d
2. (i), (ii) and (vi) are in proportion 3. $x = 4\frac{2}{3}$
4. (i) 9 (ii) $\frac{7}{11}$ (iii) 6 (iv) 21 (v) 5 (vi) 18
5. (i) 30 (ii) 24 (iii) 42 (iv) 18 (v) 60 (vi) 66

EXERCISE 6.3

1. 45 marks 2. 18 kg 3. 240 km 4. Rs 160
5. 11 : 210, 199 : 210 6. 1460 minutes 7. 39 lines 8. 48 days

REVIEW EXERCISE 6

2. (i) 4 : 6 (ii) 5 : 4 (iii) 2 : 8 (iv) 3 : 5
3. (i) 4 (ii) 7 (iii) 2 (iv) 9 (v) 144 (vi) 54
4. (ii) and (iii) are inverse proportions
5. (i) 35 : 67 (ii) 32 : 67 (iii) 32 : 35 (iv) 35 : 32
6. 25 minutes 7. Rs 1350 8. 12 workers 9. 400 persons

EXERCISE 7.1

1. (i) 20% (ii) 22% (iii) 75%
2. (i) $\frac{1}{4}$, 0.25 (ii) $\frac{1}{100}$, 0.01 (iii) $\frac{31}{100}$, 0.31 (iv) $\frac{3}{20}$, 0.15
- (v) $\frac{17}{200}$, 0.085 (vi) $\frac{41}{200}$, 0.205 (vii) $\frac{7}{4}$, 1.75 (viii) $\frac{23}{20}$, 1.15
- (ix) $\frac{9}{4}$, 2.25 (x) $\frac{1}{40}$, 0.025

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3. (i) 25% (ii) 35% (iii) 32% (iv) 125%
4. (i) 0.25 (ii) 0.31 (iii) 0.01 (iv) 0.025 (v) 0.025 (vi) 0.065
5. (i) 2.5% (ii) 40% (iii) 85% (iv) 10.5% (v) 1250% (vi) 12550%
6. (i) 70% (ii) 50% (iii) 75% (iv) 125% (v) 100%
7. $\frac{3}{10}$ 8. 64%
9. (i) 5 (ii) 21 (iii) $\frac{125}{2}$ (iv) 25 (v) $\frac{1125}{2}$ (vi) 62.5 cm
(vii) 180 litres (viii) Rs 2.65 (ix) 4.84 hours
(x) 43 metres (xi) $\frac{63}{2}$ (xii) $\frac{3033}{4}$ (xiii) $\frac{1183}{2}$ (xiv) $\frac{2107}{20}$
10. Ascending order: $\frac{1}{4}$, $\frac{2}{5}$, $\frac{3}{4}$, Descending order: $\frac{3}{4}$, $\frac{2}{5}$, $\frac{1}{4}$

EXERCISE 7.2

1. 80% 2. 60% 3. 5610 4. 15750 5. Rs 4000 6. 200 km
7. 75% 8. 20% 9. 27° , 63° 10. 110 sq. m 11. Rs 28

EXERCISE 7.3

1. (i) Profit: $\frac{25}{3}\%$ (ii) Loss: $\frac{32}{5}\%$ (iii) Profit: $\frac{50}{3}\%$
(iv) Loss: $\frac{145}{6}\%$ (v) Profit: 5.96%
2. Profit: Rs 100, $\frac{25}{6}\%$ 3. Profit: Rs 50, $\frac{25}{6}\%$
4. Loss: Rs 150 5. Rs 65000 6. Rs 1275

REVIEW EXERCISE 7

1. Profit: 160% 2. 25%
3. Imran: 68.5%, Sister: 76.5% sister performs better 4. Loss: $\frac{20}{3}\%$
5. Rs 897435.80

ANSWERS

6. (i) 5% (ii) 0.64% (iii) 0.8% (iv) 19% (v) $\frac{1}{2}$
(vi) 25% (vii) $\frac{3}{4}$ (viii) 130% (ix) $\frac{1}{10}$
7. (a) (i) 75 (ii) 62.5 (iii) $\frac{3}{10}$ (iv) 12.5
(v) 0.38 (vi) 43.2 (vii) C.P (viii) Profit
- (b) (i) False (ii) True (iii) True (iv) False
(v) False (vi) True (vii) True

EXERCISE 8.1

1. (i) $x + 2 = 8$ (ii) $7 \times y > 2$ (iii) $6 + z < 4$
2. (i) False (ii) True (iii) False (iv) True (v) False (vi) True
3. (i) Open statement (ii) Open statement (iii) Not open statement
4. (i) 9 (ii) 3 (iii) 6 (iv) 4 (v) 3 (vi) 6

EXERCISE 8.2

1. (i) 1 (ii) 1 (iii) 2 (iv) 2 (v) 3 (vi) 2
2. (i) x (ii) x (iii) x, y (iv) x, y (v) x, y (vi) y, z
(vii) x, y, z (viii) x, y, z
3. (i) 2 (ii) -3 (iii) 1 (iv) $\frac{1}{2}$ (v) 9 (vi) 10
4. (i) -7 (ii) 5 (iii) 2, 3 (iv) $1, \frac{1}{2}, -\frac{1}{4}$
(v) 5 (vi) $\frac{1}{4}, 6$
5. $2xy, 140xy, \frac{1}{5}xy, \frac{1}{4}xy, \frac{1}{3}xy, -2xy, 5xy$ are like terms,
 $4lm, 5lm$ are like terms, $-7xz, xz$ are like terms,
 $-9xyz$, and xyz are like terms and $-2p$ and $46p$ are like terms.

EXERCISE 8.3

1. (i) $7x$ (ii) $12a$ (iii) $19lm$ (iv) $16xy$

ANSWERS

2. (i) $11x$ (ii) $6a$ (iii) $20x$ (iv) $17st$
3. (i) $8a + 3b$ (ii) $10x + 9y + 8z$ (iii) $6pq + 2qr + 7pr$
(iv) $8c + 11d + 11f$
4. (i) $3x$ (ii) $12y$ (iii) $4x + 5y$ (iv) $30f - 10g - 20$
5. (i) $3x$ (ii) $21ab$ (iii) $2z$ (iv) $27xy$
(v) $5x + 2y$ (vi) $x - 5xb + 6y$

EXERCISE 8.4

1. (i) $4x + 2y$ (ii) $-2a + 3b$ (iii) $-a - 2b$
(iv) $24l + 3m + 1$ (v) $5y - 46x$
2. (i) 6 and -10 (ii) (a) 9 (b) -8 (c) -5 (iv) $\frac{33}{4}$
3. (i) 0 (ii) -23 (iii) $-\frac{7}{2}$ (iv) $-\frac{1}{10}$ (v) -2
(vi) $-\frac{1}{20}$ (vii) $-\frac{2}{3}$ (viii) 80 (ix) $\frac{1}{4}$ (x) $\frac{16}{15}$
4. 16 5. (i) Rs 5x (ii) $5x + 3$ (iii) Rs 153

REVIEW EXERCISE 8

1. (i) Rs 18x (ii) $9b + 5a$ (iii) $3x + 5$ (iv) $2y - \frac{2}{3}$ (v) $6 + p$
2. (i) Sum of x and 2 (ii) Three times y minus four
(iii) Sum of x and 5 (iv) Sum of s and twice of t
(v) Half of difference of x and y
3. (i) 2 (ii) 1 (iii) 3 (iv) 4 (v) 3
4. $5x$, $3x + 4$ and $6x + 5y + 7$
5. (i) $\frac{20}{3}xy$ (ii) $\frac{63}{5}ab$
6. (i) $8x + 13y$ (ii) $12b + 12c + 7d$ (iii) $20xy + 8x + 10y$
7. $x + 4y$ 8. $x - 18y - 14z$
9. (i) False (ii) False (iii) True
10. (i) 9 (ii) 13 (iii) 6 (iv) 3 (v) 2

ANSWERS

EXERCISE 9.1

1. $x - 20 = 32$ 2. $x + 8 = 15$ 3. $5x = 35$ 4. $\frac{16}{4} = 4$
5. $x + 2 = 4$ 6. $x - 15 = 5$ 7. $2x + 4 = 20$
8. $a = 15 + b$ where $a = \text{my age}$ $b = \text{brother's age}$

EXERCISE 9.2

1. 15 2. 7 3. 6 4. 2 5. 10
6. 18 7. 12 8. 8 9. 4 10. 6
11. -3 12. 11 13. 4 14. $\frac{26}{3}$ 15. $\frac{16}{3}$

EXERCISE 9.3

1. 19 2. 36 runs 3. 3 4. 5 5. 9
6. 1 year 7. 105 8. Rs 32

REVIEW EXERCISE 9

1. (i) $x + 4$ (ii) $x - 7$ (iii) $9x$ (iv) $\frac{4}{6}$
(v) $x + y$ (vi) $8x$
2. (i) $x - 3 = 10$ (ii) $3y + 5 = 17$ (iii) $2z + 4 = 20$ (iv) $7 - \frac{m}{2} = 4$
(v) $2x + 4 = 16$ (vi) $x + 8 = 3x$
3. (i) $\frac{21}{5}$ (ii) 9 (iii) 4 (iv) $\frac{11}{4}$ (v) 35 (iv) 6
4. 14 metres 5. 13 6. 8 7. 3 8. 16

EXERCISE 10.1

1. (i) 5 cm (ii) 8.3 cm (iii) 7.8 cm (iv) 10.0 cm
2. (i) 2 cm (ii) 1.5 cm (iii) 2.1 cm (iv) 1.8 cm (v) 1.5 cm

REVIEW EXERCISE 10

2. (i) 9.8 cm (ii) 0.8 cm (iii) 2.3 cm
6. PQ 8. (i) False (ii) False (iii) True (iv) True (v) False
9. (i) b (ii) a (iii) b (iv) c

EXERCISE 11.1

1. (a) 20 cm, 25 sq. cm (b) 114 cm, 12.25 sq. cm (c) 30 cm, 30 sq. cm
2. (a) 16 cm, 15 sq. cm (b) 17 cm, 15 sq. cm (c) 15 cm, 12.96 sq. cm
(d) 24 cm, 20 sq. cm
3. (i) 15 cm, 12.5 sq. cm (ii) 20 m, 25 sq. cm
4. (a) 12 cm (b) 12.5 cm
5. 30 cm 6. 52.7 sq. cm

EXERCISE 11.2

1. (i) 350 sq. cm (ii) 800 sq. cm (iii) 430 sq. cm
2. 550 sq. cm 3. 400 sq. cm

EXERCISE 11.3

1. 30,000 rupees 2. 21,000 rupees 3. Bismah covers more distance
4. 41 m 5. 162412.5 rupees 6. 160800 rupees 7. 325 sq. m

EXERCISE 11.4

1. (a) $\overline{AE}$, 4 cm (b) $\overline{OP}$, 4.5 cm (c) $\overline{ZS}$, 4 cm (d) $\overline{YT}$, 6 cm
2. (i) 60 sq. cm (ii) 120 sq. cm (iii) 24.75 sq. cm (iv) 30.55 sq. cm
3. 16 cm 4. 12 m 5. 3 cm

EXERCISE 11.5

1. (i) 22 sq. cm (ii) 8 sq. cm (iii) 30.25 sq. cm
2. 35 sq. cm 3. 8 cm
4. (i) 24 sq. cm (ii) 22.5 sq. cm (iii) 3.75 sq. cm (iv) 211.375 sq. cm
5. 16.5 sq. cm 6. 16 cm 7. 40 sq. cm

REVIEW EXERCISE 11

1. (i) 21.2 cm, 28.09 sq. cm (ii) 38 cm, 88 sq. cm
2. (i) 7.5 sq. cm (ii) 48 sq. cm (iii) 16.25 sq. cm
3. 244 sq. m 4. 225 5. 4 cm 6. 18 cm 7. 10 cm
8. 54 sq. cm 9. 67.5 sq. cm 10. 40 sq. cm 11. 8 cm
12. 10 cm 13. 20 sq. cm
14. (i) sq. units (ii) Units (iii) Product (iv) Sum
 (v) Length, breadth (vi) 4, length of side (vii) Perimeter = $2(l + b)$
 (viii) Area = Base $\times$ height (ix) Area = $\frac{1}{2}$ (Base $\times$ height)
 (x) Area = $\frac{h}{2}(b_1 + b_2)$

EXERCISE 12.1

1. (i) sphere (ii) cylinder (iii) cone (iv) cube (v) cuboid (vi) cuboid
2. (i) (a) 12 (b) 6 (c) 8 (ii) (a) vertex (b) 1
 (iii) (a) edge (b) vertex (c) 6

EXERCISE 12.2

1. (i) sq. cm, cubic cm (ii) sq. mm, cubic mm
2. (i) 24 sq. cm, 8 cubic cm (ii) 86.5 sq. cm, 48.75 cubic cm
 (iii) 73.5 sq. cm, 42.875 cubic cm (iv) 48 sq. cm, 18 cubic cm
3. (i) 125 cubic cm (ii) 238.328 cubic cm
 (iii) 15.625 cubic cm

EXERCISE 12.3

1. Rs 11000
2. 50 cubic m
3. 1125000 cubic cm
4. Rs 1875
5. 125 cubic m
6. 8
7. Rs 1800
8. 65.45 cubic m
9. 19250 cubic cm
10. Rs 2514.72

REVIEW EXERCISE 12

1. (i) 8 (ii) 1 (iii) 6 (iv) 2 (v) 12
2. Tennis ball, Geometry box and Brick
3. (i) 37.5 sq. cm, 15.625 cubic cm (ii) 30.5 sq. cm, 10.5 cubic cm
4. 13.5 sq. cm and 3.375 cubic cm
5. Rs 21400
6. 175.5 cubic m

EXERCISE 13.1

3. (ii), (iii)

EXERCISE 13.2

5. (i) Tennis (ii) Cricket (iii) Hockey and football
6. (i) Science (ii) English (iii) Science
7. (i) Class II (ii) Class V (iii) Class III and IV

REVIEW EXERCISE 13

1. (i) Maths, 79 marks (ii) Social Studies, 40 marks
(iii) Urdu and Sindhi, 50 marks each
(iv)

Subject	Sindhi	Urdu	English	Science	Social Studies	Islamiat	Maths
Marks	50	50	60	70	40	78	79

2. (i) True (ii) True (iii) False (iv) False
3. (i) (ii) (iii) Vertically (iv) Data (v) Data